

Unit 14: Equations

1. Mrs. Dunbar placed a new 3-gallon water bottle in the office water dispenser when she arrived in the morning. At the end of the day, the bottle had $1\frac{3}{8}$ gallons remaining. The equation $3 = x + 1\frac{3}{8}$ can be used to find x , the gallons of water dispensed from the bottle. How much water was dispensed from the bottle?

$$\begin{array}{r} \frac{3}{1} - \frac{11}{8} \\ \downarrow \\ \frac{24}{8} - \frac{11}{8} = \frac{13}{8} \end{array}$$

$$\begin{array}{r} 3 = x + 1\frac{3}{8} \\ -1\frac{3}{8} \quad -1\frac{3}{8} \\ \hline 1\frac{5}{8} = x \end{array}$$

2. Solve the equations

a. $21 = m - 15$

$$\begin{array}{r} +15 \quad +15 \\ \hline 36 = m \end{array}$$

b. $d + 66 = -18$

$$\begin{array}{r} -66 \quad -66 \\ \hline d = -84 \end{array}$$

c. $h - 20 = -18$

$$\begin{array}{r} +20 \quad +20 \\ \hline h = 2 \end{array}$$

3. Which of the expressions are equivalent?

a. $5 \cdot m + 5 \cdot 7$

$$5m + 35 \checkmark$$

b. $5(m + 7)$

$$\begin{array}{l} 5 \cdot m = 5m \\ 5 \cdot 7 = 35 \end{array}$$

$$5m + 35 \checkmark$$

c. $5m + 7$

$$5m + 7$$

4. Lilly has x flowers. Gill has 8 times more flowers than Lilly. How many flowers does Gill have?

$$8x$$

5. Josephine told her brother she would give him $\frac{1}{3}$ of the money they made at a lemonade stand. She gave her brother \$24.25. This situation can be represented by the equation $\frac{1}{3}x = 24.25$. Find x , the amount of money they made at the lemonade stand.

$$3\left(\frac{1}{3}x\right) = (24.25)3$$

$$x = 72.75$$